

Financial Data Import, Transformation & Time Intelligence

INTRODUCTION:

This project focuses on analyzing Shopify stock market data using Microsoft Power BI. The dataset contains historical stock information such as opening price, closing price, highest price, lowest price, adjusted closing price and trading volume.

Power BI was used to import and prepare the stock data, create a dedicated Date Table, establish relationships between tables, develop DAX measures and build an interactive dashboard.

The main purpose of the project is to understand how Shopify's stock prices and trading activity change over time.

OBJECTIVES:

The main objectives of this project are:

- To import and prepare Shopify stock data in Power BI.
- To create a dedicated Date Table using DAX.
- To establish a relationship between the Date Table and stock data.
- To analyze stock prices using time-based calculations.
- To create Year, Quarter, Month and Day fields.
- To visualize Shopify stock performance through an interactive dashboard.

DATASET DESCRIPTION:

The dataset used for this project is Shopify Stock Data.

The main columns used in the analysis are:

- date – Date of the stock data
- open – Opening stock price
- high – Highest stock price
- low – Lowest stock price
- close – Closing stock price
- adj_close – Adjusted closing price
- volume – Trading volume

DATA IMPORT AND DESCRIPTION:

The Shopify stock dataset was imported into **Power BI Desktop**.

After importing the dataset, the stock data table was made available in the Power BI data model.

The important fields used in the project were:

- Date
- Open Price
- High Price
- Low Price
- Close Price
- Adjusted Close Price
- Trading Volume

The dataset was prepared so that the stock price and trading activity could be analyzed using Power BI visuals.

DATE TABLE:

A separate Date Table was created to support time-based analysis.

The Date Table contains the following fields:

- Date
- Day
- Month
- Month Number
- Quarter
- Year

The Date Table provides a structured way to analyze stock prices across different years, quarters and months.

A relationship was established between the Date Table and the main Shopify stock data table.

DATA MODEL AND RELATIONSHIP:

A relationship was created between the Date Table and the Shopify stock data table.

The relationship is:

Date Table[Date] → SHOP...[date]

The relationship uses a one-to-many (1:*) cardinality.

- Date Table is on the one side.
- Shopify stock data is on the many side.

This relationship allows the Date Table to filter and organize the stock data for time-based analysis.

FINANCIAL DAX MEASURES:

Several DAX measures were created to analyze the Shopify stock data.

The measures created in the project include:

Close Price

This measure is used to calculate the closing stock price.

Open Price

This measure represents the opening stock price.

High Price

This measure represents the highest stock price.

Low Price

This measure represents the lowest stock price.

Trading Volume

This measure is used to analyze the trading volume of the stock.

Daily Return %

This measure is used to analyze the percentage change between the opening and closing prices.

These measures were organized inside the Measures Table.

KPI CARDS:

The KPI cards display important stock market values in an easy-to-understand format.

Open Price

Displays the opening price of the stock.

Close Price

Displays the closing price of the stock.

High Price

Displays the highest recorded price.

Low Price

Displays the lowest recorded price.

Trading Volume

Displays the number of shares traded.

Daily Return %

Displays the percentage change between opening and closing prices.

DASHBOARD VISUALIZATIONS:

1.Close Price by Year

A line chart was created using Year on the X-axis and Close Price on the Y-axis. This visualization shows how the closing stock price changed across different years.

2. High and Low Price Trend

A line chart was created using the Date field and the High Price and Low Price measures. This helps compare the highest and lowest stock prices over time.

3. Trading Volume Over Time

A column chart was created using Date and Trading Volume. This visualization shows changes in trading activity over time.

4. Monthly Close Price

A line chart was created using Month and Close Price. The Month Number field was used to maintain the correct January-to-December order.

5. Previous Year Comparison

The previous-year measures can be used to compare current closing prices with corresponding previous-year values.

Interactive Features:

The dashboard includes interactive filtering using the Date Table.

A Year slicer can be used to select a particular year.

A Month slicer can be used to filter the dashboard based on a particular month.

When a year or month is selected, the related cards and charts update according to the selected period.

DASHBOARD:

The completed Power BI dashboard provides a visual summary of Shopify stock performance.

It combines:

- KPI cards
- Stock price charts
- Trading volume analysis
- Time-based analysis
- Previous-year comparison
- Interactive slicers

CONCLUSION:

The Shopify Stock Market Analysis project was successfully developed using Microsoft Power BI. The Shopify stock dataset was imported and prepared for analysis, and a separate Date Table was created to support time-based analysis.

DAX measures were created for important stock indicators such as Open Price, Close Price, High Price, Low Price, Trading Volume and Daily Return %. A relationship between the Date Table and stock data was established to support filtering and time-based analysis.

Finally, an interactive dashboard was created using KPI cards, charts and slicers. The dashboard provides a simple visual representation of Shopify stock prices and trading activity over time.

This project helped demonstrate the use of Power BI, data modeling, DAX measures, date analysis and interactive data visualization for financial data.